

RELATORIO DE AVALIACAO E PREPARACAO - RENUMF90

Data de Processamento: 01/05/2026 as 06:18:10 ate 01/05/2026 as 06:18:24

1. CONFIGURACAO DO MODELO BIOLOGICO

Caracteristicas (Traits): PN_kg, PD_kg, PS_kg, GPD_g-dia, PE_cm, AOL_cm2, EGS_mm, MAR_%, CAR_kg-dia, PREC_SEX, IPP_dias, PROB_3P_%, STAY_%

Efeitos Fixos: GC, Fazenda, Estacao, Lote_Manejo, Regime_Alimentacao, Tipo_Pastagem, Localidade_Bloco, Piquete, Sexo, Geracao, Raca

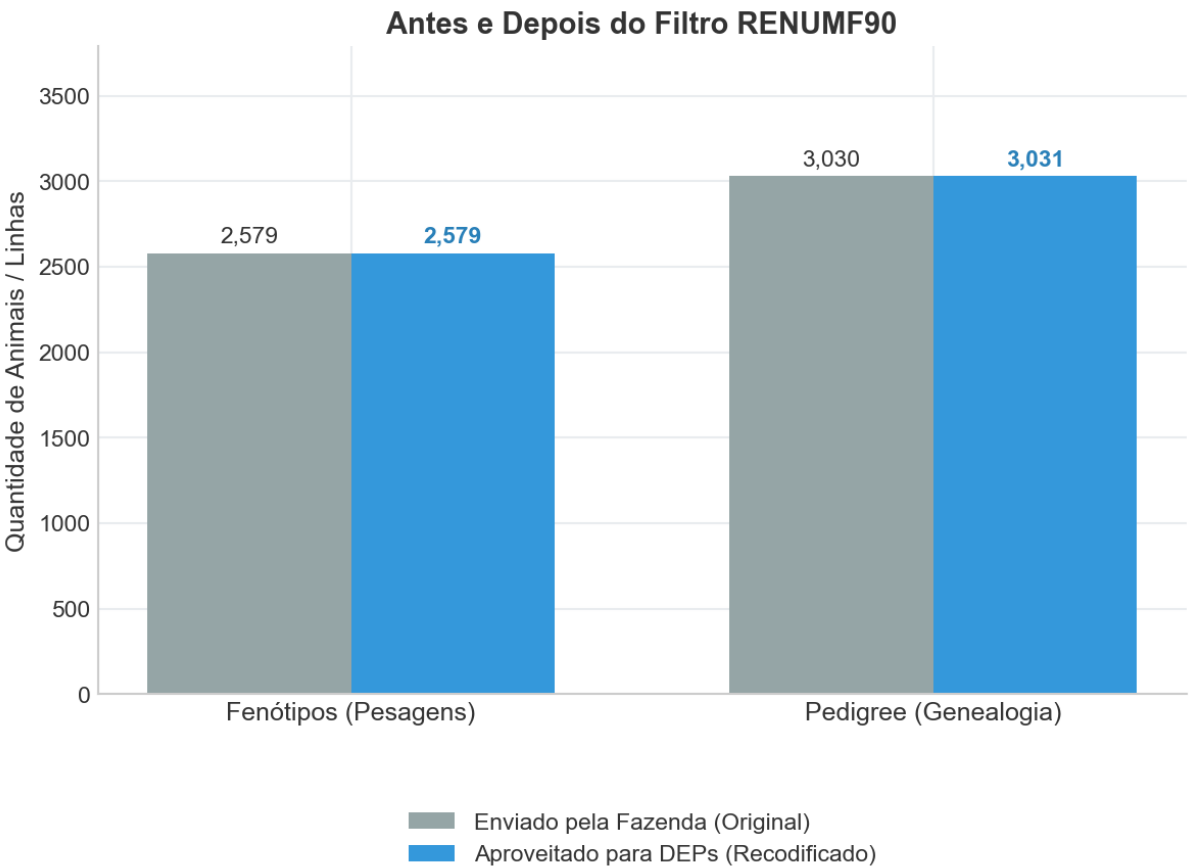
Covariaveis: itu_media, itu_dp, itu_max, itu_min, itu_leituras_criticas

2. RETENCAO E AUDITORIA DE DADOS

FENOTIPOS: Fornecidos (2579) -> Aproveitados (2579)

PEDIGREE: Fornecidos (3030) -> Aproveitados (3031)

GENOMICA: Total Analisado (3030) -> Isolados/Inuteis (0)



3. SAUDE GENETICA E ALERTAS

Consanguinidade Max: N/D | Media: N/D

4. LOG TECNICO ORIGINAL (MOTOR FORTRAN)

```
=====
[PIPELINE LINEAR]
=====
RENUMF90 version 1.168
* START job      Wall time: 05-01-2026  06h 18m 11s 853
name of parameter file? renum_linear.par
datafile:fenotipos_raw.txt
traits:          9          10          11          13          14
R
  1.000      0.000      0.000      0.000      0.000
  0.000      1.000      0.000      0.000      0.000
  0.000      0.000      1.000      0.000      0.000
  0.000      0.000      0.000      1.000      0.000
  0.000      0.000      0.000      0.000      1.000

Processing effect  1 of type cross
item_kind=alpha

Processing effect  2 of type cross
item_kind=alpha

Processing effect  3 of type cross
item_kind=alpha

Processing effect  4 of type cross
item_kind=alpha

Processing effect  5 of type cross
item_kind=alpha

Processing effect  6 of type cross
item_kind=alpha

Processing effect  7 of type cross
item_kind=alpha

Processing effect  8 of type cross
item_kind=alpha

Processing effect  9 of type cross
item_kind=alpha

Processing effect 10 of type cross
item_kind=alpha

Processing effect 11 of type cross
item_kind=alpha

Processing effect 12 of type cov

Processing effect 13 of type cov

Processing effect 14 of type cov

Processing effect 15 of type cov

Processing effect 16 of type cov

Processing effect 17 of type cross
item_kind=alpha
pedigree file name  "pedigree_raw.txt"
positions of animal, sire, dam, alternate dam, yob, and group  1    2    3    0    0    0    0
SNP file name  "genotipos_raw.txt"
```

Maximum size of character fields: 20

Maximum size of record (max_string_readline): 8000

Maximum number of fields for input file (max_field_readline): 100

Pedigree search method (ped_search): convention

Order of pedigree animals (animal_order): default

Order of UPG (upg_order): default

Missing observation code (missing): -999

guessed number of lines 2579

Using prime hash function

hash tables for effects set up

first 3 lines of the data file (up to 700 characters)

1 122 2 2 3 3 2 3 4.2 3.5 -0.5 0 -999 28.21 0 1 1 1 4 1 1 20240322 2 114.3 75.37 6.82 87.31 -0.2056672425865524 133 1 1 1
20200101

2 105 4 4 4 4 3 4 5.1 4.2 0.12 0 -999 72.98 0 1 1 2 1 2 1 20240226 38 114.3 77.56 5.64 86.99 1.5411790583507272 145 1 1 1
20200101

3 141 1 1 1 2 1 1 3.8 3.1 -0.85 0 -999 83.38 0 1 2 3 1 3 1 20240611 41 114.3 68.62 5.72 78.72 -0.5160008312446829 72 1 1 1
20200101

% records read: 38.7747%, records read: 1000 timing 0.04687500

% records read: 77.5494%, records read: 2000 timing 0.09375000

read 2579 records in 0.1093750 seconds

Processing effect group 1

table with 108 elements sorted

added count

Effect group 1 of column 1 with 108 levels

table expanded from 10000 to 10000 records

Effect group 1 done in 0.6406250 seconds

Processing effect group 2

table with 4 elements sorted

added count

Effect group 2 of column 1 with 4 levels

table expanded from 10000 to 10000 records

Effect group 2 done in 0.3437500 seconds

Processing effect group 3

table with 5 elements sorted

added count

Effect group 3 of column 1 with 5 levels

table expanded from 10000 to 10000 records

Effect group 3 done in 0.3750000 seconds

Processing effect group 4

table with 13 elements sorted

added count

Effect group 4 of column 1 with 13 levels

table expanded from 10000 to 10000 records

Effect group 4 done in 0.4375000 seconds

Processing effect group 5

table with 4 elements sorted

added count

Effect group 5 of column 1 with 4 levels

table expanded from 10000 to 10000 records

Effect group 5 done in 0.3906250 seconds

Processing effect group 6

```

table with          5 elements sorted
added count
Effect group        6 of column          1 with          5 levels
table expanded from 10000 to          10000 records
Effect group        6 done in  0.3593750    seconds
-----
Processing effect group  7
table with          10 elements sorted
added count
Effect group        7 of column          1 with          10 levels
table expanded from 10000 to          10000 records
Effect group        7 done in  0.5312500    seconds
-----
Processing effect group  8
table with          161 elements sorted
added count
Effect group        8 of column          1 with          161 levels
table expanded from 10000 to          10000 records
Effect group        8 done in  0.3906250    seconds
-----
Processing effect group  9
table with          2 elements sorted
added count
Effect group        9 of column          1 with          2 levels
table expanded from 10000 to          10000 records
Effect group        9 done in  0.3593750    seconds
-----
Processing effect group 10
table with          1 elements sorted
added count
Effect group       10 of column          1 with          1 levels
table expanded from 10000 to          10000 records
Effect group       10 done in  0.3437500    seconds
-----
Processing effect group 11
table with          1 elements sorted
added count
Effect group       11 of column          1 with          1 levels
table expanded from 10000 to          10000 records
Effect group       11 done in  0.3437500    seconds
-----
Processing effect group 12
-----
Processing effect group 13
-----
Processing effect group 14
-----
Processing effect group 15
-----
Processing effect group 16
-----
Processing effect group 17
added count
Effect group       17 of column          1 with          2579 levels
Effect group       17 done in  0.0000000E+00 seconds
wrote statistics in file "renf90.tables"
reading data file for basic statistics
% records read:    38.7747%, records read:    1000 timing    0.04687500
% records read:    77.5494%, records read:    2000 timing    0.07812500

Basic statistics for input data (missing value code is '-999')
Pos  Min      Max      Mean      SD          N
  9   1.7600    6.8500    4.3522    0.74329    2579
 10   1.7500    5.1400    3.5173    0.48949    2579
 11  -1.0700    0.85000   -0.12120   0.33410    2579
 13   848.00   1072.0    988.57    56.053     30

```

14	20.030	85.000	52.870	18.769	2579
25	-999.00	77.960	57.430	114.76	2579
26	-999.00	7.3200	-6.1178	107.74	2579
27	-999.00	90.210	67.240	115.86	2579
28	-999.00	1.8847	-11.611	107.14	2579
29	-999.00	160.00	84.398	138.77	1990

Correlation matrix

	9	10	11	13	14	25	26	27	28	29
9	1.00	0.06	-0.01	0.21	0.01	-0.00	-0.00	-0.00	-0.00	-0.01
10	0.06	1.00	0.02	0.14	0.03	-0.00	-0.00	-0.00	-0.00	0.00
11	-0.01	0.02	1.00	-0.37	-0.00	-0.04	-0.04	-0.04	-0.04	-0.03
13	0.21	0.14	-0.37	1.00	0.01	-0.07	-0.07	-0.07	-0.07	-0.31
14	0.01	0.03	-0.00	0.01	1.00	0.01	0.01	0.01	0.01	0.00
25	-0.00	-0.00	-0.04	-0.07	0.01	1.00	1.00	1.00	1.00	0.97
26	-0.00	-0.00	-0.04	-0.07	0.01	1.00	1.00	1.00	1.00	0.97
27	-0.00	-0.00	-0.04	-0.07	0.01	1.00	1.00	1.00	1.00	0.97
28	-0.00	-0.00	-0.04	-0.07	0.01	1.00	1.00	1.00	1.00	0.97
29	-0.01	0.00	-0.03	-0.31	0.00	0.97	0.97	0.97	0.97	1.00

Counts of nonzero values (order as above)

2579	2579	2579	30	2579	2579	2579	2579	2579	1990
2579	2579	2579	30	2579	2579	2579	2579	2579	1990
2579	2579	2579	30	2579	2579	2579	2579	2579	1990
30	30	30	30	30	30	30	30	30	22
2579	2579	2579	30	2579	2579	2579	2579	2579	1990
2579	2579	2579	30	2579	2579	2579	2579	2579	1990
2579	2579	2579	30	2579	2579	2579	2579	2579	1990
2579	2579	2579	30	2579	2579	2579	2579	2579	1990
2579	2579	2579	30	2579	2579	2579	2579	2579	1990
2579	2579	2579	30	2579	2579	2579	2579	2579	1990
1990	1990	1990	22	1990	1990	1990	1990	1990	1990

basic_statistics done in 0.1093750 seconds

random effect with SNPs 17

type: animal

file: genotipos_raw.txt

read SNPs 3030 records

Effect group 17 of column 1 with 3031 levels

add_snp done in 4.6875000E-02 seconds

reorder_animal done in 0.0000000E+00 seconds

random effect 17

type:animal

renumber_animal, start Wall time: 05-01-2026 06h 18m 17s 591

opened output pedigree file "renadd17.ped"

first 3 lines of the file (up to 700 characters)

```

1 3012 4055
2 3005 4210
3 3044 4102

```

load_animals time 9.3750000E-02 seconds

read 3030 pedigree records

loaded 0 parent(s) in round 1

load_animals time 4.6875000E-02 seconds

read 3030 pedigree records

Pedigree checks

compute_inbreeding time, start Wall time: 05-01-2026 06h 18m 17s 736

Computations for inbreeding coefficients

actual inbreeding computation, start Wall time: 05-01-2026 06h 18m 17s 738

actual inbreeding computation, done Wall time: 05-01-2026 06h 18m 17s 739

Tiny negative value will be replaced with 0 considered as numerical error.

Wrote inbreeding file "renf90.inb" with original id

Inbreeding statistics:

the maximum inbreeding coefficient = 0.0000

```

average inbreeding for inbred animals =      NaN    n = 0
              for all animals = 0.0000    n = 3031
compute_inbreeding time, done      Wall time: 05-01-2026 06h 18m 17s 790
loop writing renadd, time 7.8125000E-02 seconds

```

```

Number of animals with records      =      2579
Number of animals with genotypes    =      3030
Number of animals with records or genotypes =      3031
Number of animals with genotypes and no records =      452
Number of parents without records or genotypes =      0
Total number of animals              =      3031
renumber_animal, end      Wall time: 05-01-2026 06h 18m 17s 861

```

```

Wrote cross reference IDs for SNP file "genotipos_raw.txt_XrefID"
write_SNP_Xref done in 0.1093750 seconds
write_map done in 0.0000000E+00 seconds

```

```

Wrote parameter file "renf90.par"
write renf90.par done in 0.0000000E+00 seconds
writing renf90.dat
% records written: 38.7747%, records written: 1000 timing 0.14062500
% records written: 77.5494%, records written: 2000 timing 0.28125000
Wrote renumbered data "renf90.dat" 2579 records
write renf90.dat done in 0.3437500 seconds

```

```

Number of records NOT present in pedigree file: 1
Wrote field information "renf90.fields" for 22 fields in data
save_renum_table done in 0.0000000E+00 seconds
* END job      Wall time: 05-01-2026 06h 18m 18s 338

```

```

=====
[PIPELINE CATEGÓRICO (LIMIAR)]
=====

```

```

RENUMF90 version 1.168
* START job      Wall time: 05-01-2026 06h 18m 18s 399
name of parameter file? renum_categ.par
datafile:fenotipos_raw.txt
traits:          3          4          5          6          7          8
              12          15
R
  1.000      0.000      0.000      0.000      0.000      0.000      0.000
  0.000
  0.000      1.000      0.000      0.000      0.000      0.000      0.000
  0.000
  0.000      0.000      1.000      0.000      0.000      0.000      0.000
  0.000
  0.000      0.000      0.000      1.000      0.000      0.000      0.000
  0.000
  0.000      0.000      0.000      0.000      1.000      0.000      0.000
  0.000
  0.000      0.000      0.000      0.000      0.000      1.000      0.000
  0.000
  0.000      0.000      0.000      0.000      0.000      0.000      1.000
  0.000
  0.000      0.000      0.000      0.000      0.000      0.000      0.000
  1.000

```

```

Processing effect 1 of type cross
item_kind=alpha

```

```

Processing effect 2 of type cross
item_kind=alpha

```

```

Processing effect 3 of type cross
item_kind=alpha

```

Processing effect 4 of type cross
item_kind=alpha

Processing effect 5 of type cross
item_kind=alpha

Processing effect 6 of type cross
item_kind=alpha

Processing effect 7 of type cross
item_kind=alpha

Processing effect 8 of type cross
item_kind=alpha

Processing effect 9 of type cross
item_kind=alpha

Processing effect 10 of type cross
item_kind=alpha

Processing effect 11 of type cross
item_kind=alpha

Processing effect 12 of type cov

Processing effect 13 of type cov

Processing effect 14 of type cov

Processing effect 15 of type cov

Processing effect 16 of type cov

Processing effect 17 of type cross
item_kind=alpha
pedigree file name "pedigree_raw.txt"
positions of animal, sire, dam, alternate dam, yob, and group 1 2 3 0 0 0 0
SNP file name "genotipos_raw.txt"

Maximum size of character fields: 20

Maximum size of record (max_string_readline): 8000

Maximum number of fields for input file (max_field_readline): 100

Pedigree search method (ped_search): convention

Order of pedigree animals (animal_order): default

Order of UPG (upg_order): default

Missing observation code (missing): -999

guessed number of lines 2579

Using prime hash function
hash tables for effects set up
first 3 lines of the data file (up to 700 characters)

1 122 2 2 3 3 2 3 4.2 3.5 -0.5 0 -999 28.21 0 1 1 1 4 1 1 20240322 2 114.3 75.37 6.82 87.31 -0.2056672425865524 133 1 1 1
20200101
2 105 4 4 4 4 3 4 5.1 4.2 0.12 0 -999 72.98 0 1 1 2 1 2 1 20240226 38 114.3 77.56 5.64 86.99 1.5411790583507272 145 1 1 1
20200101
3 141 1 1 1 2 1 1 3.8 3.1 -0.85 0 -999 83.38 0 1 2 3 1 3 1 20240611 41 114.3 68.62 5.72 78.72 -0.5160008312446829 72 1 1 1
20200101

% records read: 38.7747%, records read: 1000 timing 0.04687500
% records read: 77.5494%, records read: 2000 timing 0.09375000
read 2579 records in 0.1250000 seconds

Processing effect group 1
table with 108 elements sorted
added count
Effect group 1 of column 1 with 108 levels
table expanded from 10000 to 10000 records
Effect group 1 done in 0.4062500 seconds

Processing effect group 2
table with 4 elements sorted
added count
Effect group 2 of column 1 with 4 levels
table expanded from 10000 to 10000 records
Effect group 2 done in 0.3906250 seconds

Processing effect group 3
table with 5 elements sorted
added count
Effect group 3 of column 1 with 5 levels
table expanded from 10000 to 10000 records
Effect group 3 done in 0.5156250 seconds

Processing effect group 4
table with 13 elements sorted
added count
Effect group 4 of column 1 with 13 levels
table expanded from 10000 to 10000 records
Effect group 4 done in 0.3906250 seconds

Processing effect group 5
table with 4 elements sorted
added count
Effect group 5 of column 1 with 4 levels
table expanded from 10000 to 10000 records
Effect group 5 done in 0.3593750 seconds

Processing effect group 6
table with 5 elements sorted
added count
Effect group 6 of column 1 with 5 levels
table expanded from 10000 to 10000 records
Effect group 6 done in 0.3906250 seconds

Processing effect group 7
table with 10 elements sorted
added count
Effect group 7 of column 1 with 10 levels
table expanded from 10000 to 10000 records
Effect group 7 done in 0.3906250 seconds

Processing effect group 8
table with 161 elements sorted
added count
Effect group 8 of column 1 with 161 levels
table expanded from 10000 to 10000 records
Effect group 8 done in 0.3906250 seconds

Processing effect group 9
table with 2 elements sorted
added count
Effect group 9 of column 1 with 2 levels
table expanded from 10000 to 10000 records
Effect group 9 done in 0.5000000 seconds


```

-----
Processing effect group 10
table with          1 elements sorted
added count
Effect group        10 of column          1 with          1 levels
table expanded from    10000 to    10000 records
Effect group        10 done in  0.3906250    seconds
-----
Processing effect group 11
table with          1 elements sorted
added count
Effect group        11 of column          1 with          1 levels
table expanded from    10000 to    10000 records
Effect group        11 done in  0.3593750    seconds
-----
Processing effect group 12
-----
Processing effect group 13
-----
Processing effect group 14
-----
Processing effect group 15
-----
Processing effect group 16
-----
Processing effect group 17
added count
Effect group        17 of column          1 with        2579 levels
Effect group        17 done in  0.0000000E+00 seconds
wrote statistics in file "renf90.tables"
reading data file for basic statistics
% records read:      38.7747%, records read:      1000 timing      0.06250000
% records read:      77.5494%, records read:      2000 timing      0.10937500

```

Basic statistics for input data (missing value code is '-999')

Pos	Min	Max	Mean	SD	N
3	1.0000	4.0000	2.2687	1.0527	2579
4	1.0000	4.0000	2.4781	1.1199	2579
5	1.0000	4.0000	2.4897	1.1163	2579
6	1.0000	4.0000	2.4843	1.1152	2579
7	1.0000	4.0000	2.2762	1.0243	2549
8	1.0000	4.0000	2.4986	1.1191	2579
12	0.0000	1.0000	0.58162E-02	0.76057E-01	2579
15	0.0000	1.0000	0.77549E-02	0.87737E-01	2579
25	-999.00	77.960	57.430	114.76	2579
26	-999.00	7.3200	-6.1178	107.74	2579
27	-999.00	90.210	67.240	115.86	2579
28	-999.00	1.8847	-11.611	107.14	2579
29	-999.00	160.00	84.398	138.77	1990

Correlation matrix

	3	4	5	6	7	8	12	15	25	26	27	28	29
3	1.00	0.16	0.16	0.10	0.22	0.13	-0.02	-0.03	0.03	0.03	0.03	0.03	0.03
4	0.16	1.00	0.16	-0.16	0.23	0.14	-0.03	-0.05	0.05	0.05	0.05	0.05	0.05
5	0.16	0.16	1.00	0.87	0.12	0.12	0.00	-0.00	-0.05	-0.05	-0.05	-0.05	-0.06
6	0.10	-0.16	0.87	1.00	0.04	0.06	-0.04	-0.05	0.05	0.05	0.05	0.05	0.05
7	0.22	0.23	0.12	0.04	1.00	0.14	NaN	NaN	0.07	0.07	0.06	0.07	0.07
8	0.13	0.14	0.12	0.06	0.14	1.00	0.00	-0.00	-0.05	-0.05	-0.05	-0.05	-0.05
12	-0.02	-0.03	0.00	-0.04	NaN	0.00	1.00	0.63	-0.47	-0.47	-0.47	-0.47	-0.53
15	-0.03	-0.05	-0.00	-0.05	NaN	-0.00	0.63	1.00	-0.57	-0.57	-0.57	-0.57	-0.62
25	0.03	0.05	-0.05	0.05	0.07	-0.05	-0.47	-0.57	1.00	1.00	1.00	1.00	0.97
26	0.03	0.05	-0.05	0.05	0.07	-0.05	-0.47	-0.57	1.00	1.00	1.00	1.00	0.97
27	0.03	0.05	-0.05	0.05	0.06	-0.05	-0.47	-0.57	1.00	1.00	1.00	1.00	0.97
28	0.03	0.05	-0.05	0.05	0.07	-0.05	-0.47	-0.57	1.00	1.00	1.00	1.00	0.97
29	0.03	0.05	-0.06	0.05	0.07	-0.05	-0.53	-0.62	0.97	0.97	0.97	0.97	1.00

```

Counts of nonzero values (order as above)
    2579    2579    2579    2579    2549    2579    2579    2579    2579    2579    2579    2579
1990
    2579    2579    2579    2579    2549    2579    2579    2579    2579    2579    2579    2579
1990
    2579    2579    2579    2579    2549    2579    2579    2579    2579    2579    2579    2579
1990
    2579    2579    2579    2579    2549    2579    2579    2579    2579    2579    2579    2579
1990
    2549    2549    2549    2549    2549    2549    2549    2549    2549    2549    2549    2549
1968
    2579    2579    2579    2579    2549    2579    2579    2579    2579    2579    2579    2579
1990
    2579    2579    2579    2579    2549    2579    2579    2579    2579    2579    2579    2579
1990
    2579    2579    2579    2579    2549    2579    2579    2579    2579    2579    2579    2579
1990
    2579    2579    2579    2579    2549    2579    2579    2579    2579    2579    2579    2579
1990
    2579    2579    2579    2579    2549    2579    2579    2579    2579    2579    2579    2579
1990
    2579    2579    2579    2579    2549    2579    2579    2579    2579    2579    2579    2579
1990
    2579    2579    2579    2579    2549    2579    2579    2579    2579    2579    2579    2579
1990
    1990    1990    1990    1990    1968    1990    1990    1990    1990    1990    1990    1990
1990
basic_statistics done in  0.1250000    seconds

random effect with SNPs 17
type: animal
file: genotipos_raw.txt
read SNPs      3030 records
Effect group   17 of column      1 with      3031 levels
add_snp done in  4.6875000E-02 seconds
reorder_animal done in  0.0000000E+00 seconds

random effect  17
type:animal
renumber_animal, start   Wall time: 05-01-2026  06h 18m 24s 165
opened output pedigree file "renadd17.ped"
first 3 lines of the file (up to 700 characters)
  1 3012 4055
  2 3005 4210
  3 3044 4102
load_animals time  9.3750000E-02 seconds
read      3030 pedigree records
loaded      0 parent(s) in round      1
load_animals time  4.6875000E-02 seconds
read      3030 pedigree records

Pedigree checks
compute_inbreeding time, start   Wall time: 05-01-2026  06h 18m 24s 303

Computations for inbreeding coefficients
actual inbreeding computation, start   Wall time: 05-01-2026  06h 18m 24s 305
actual inbreeding computation, done    Wall time: 05-01-2026  06h 18m 24s 306
Tiny negative value will be replaced with 0 considered as numerical error.
Wrote inbreeding file "renf90.inb" with original id

Inbreeding statistics:
the maximum inbreeding coefficient    =  0.0000
average inbreeding for inbred animals =    NaN   n = 0
      for all animals =  0.0000   n = 3031
compute_inbreeding time, done    Wall time: 05-01-2026  06h 18m 24s 369
loop writing renadd, time  6.2500000E-02 seconds

```

```
Number of animals with records          =          2579
Number of animals with genotypes         =          3030
Number of animals with records or genotypes =          3031
Number of animals with genotypes and no records =          452
Number of parents without records or genotypes =           0
Total number of animals                  =          3031
renumber_animal, end      Wall time: 05-01-2026  06h 18m 24s 438
```

```
Wrote cross reference IDs for SNP file "genotipos_raw.txt_XrefID"
write_SNP_Xref done in   0.1093750      seconds
write_map done in   0.0000000E+00  seconds
```

```
Wrote parameter file "renf90.par"
write renf90.par done in   1.5625000E-02  seconds
writing renf90.dat
% records written:      38.7747%, records written:      1000 timing      0.09375000
% records written:      77.5494%, records written:      2000 timing      0.17187500
Wrote renumbered data "renf90.dat" 2579 records
write renf90.dat done in   0.2187500      seconds
```

```
Number of records NOT present in pedigree file: 1
Wrote field information "renf90.fields" for 25 fields in data
save_renum_table done in   0.0000000E+00  seconds
* END job      Wall time: 05-01-2026  06h 18m 24s 782
```